

RIPHAH INTERNATIONAL COLLEGE (RIC)

Harbanspura Campus, Lahore, Pakistan

DATABASE SYSTEMS

Practical & Research Based Assignment

Program:	ADP (Associate Degree Programme)
Subject:	Database Systems
Campus:	RIC Harbanspura Campus, Lahore
Instructor:	Hafiz Muhammad Usman
Total Marks:	40
Submission Date:	22 – May – 2026 (Saturday)
Submission Mode:	Printed Hard Copy (Handwritten or Typed)

■ **PLAGIARISM WARNING:** If more than 20% of your assignment content is copied from any AI tool (ChatGPT, Gemini, etc.) or the internet, your marks will be graded as ZERO (0). Plagiarism of any kind is strictly NOT allowed.

General Instructions:

1. This is a **practical and research-based** assignment. Each question requires you to write real SQL queries and provide screenshots of their output.
2. Create a database named **RIC_StudentDB** in MySQL/MariaDB or any RDBMS of your choice (must match screenshots).
3. Every question must include: **(a) SQL Query** written clearly, and **(b) Screenshot** of the query execution and result.
4. Submit in **printed format**. Screenshots must be pasted and printed – not handdrawn.
5. Write your **Name, Roll No., and Section** on the first page.
6. Late submissions will NOT be accepted.

■ ASSIGNMENT SCENARIO

You are a Database Administrator (DBA) at **City General Hospital, Lahore**. The hospital management has asked you to design and implement a small database system to manage **Patients, Doctors, and Appointments**. You must create the database, populate it with realistic data, and perform various SQL operations as described below. All work must be demonstrated with screenshots from your SQL environment (MySQL Workbench, phpMyAdmin, or command line).

Table	Columns
patients	patient_id (PK), full_name, age, gender, city, contact_no, admitted_date
doctors	doctor_id (PK), full_name, specialization, experience_years, fee
appointments	appointment_id (PK), patient_id (FK), doctor_id (FK), appt_date, status, diagnosis

PART A: INSERT Operations [10 Marks]

Q1. Database & Table Creation [4 Marks]

Create the database **RIC_HospitalDB** and all three tables with appropriate data types, primary keys, and foreign key constraints.

Required Deliverables:

- Write the full CREATE DATABASE and CREATE TABLE SQL statements.
- Take a screenshot showing the tables listed in your database (SHOW TABLES;).
- Screenshot of each table structure (DESCRIBE tablename; or equivalent).

Q2. Data Insertion – Hospital Records [6 Marks]

Insert the following **minimum records** into your tables:

- At least **6 patients** (include variety in cities, ages, genders).
- At least **4 doctors** (different specializations e.g. Cardiologist, Neurologist, Orthopedic, General Physician).
- At least **8 appointments** (link different patients with different doctors; use different dates and statuses such as Scheduled, Completed, Cancelled).

Required Deliverables:

- Write all INSERT INTO statements used.
- Screenshot of SELECT * FROM each table showing all inserted data.
- Explain in 2–3 lines: What happens if you try to insert an appointment with a non-existent patient_id? Why?

PART B: SELECT Queries [12 Marks]

Q3. Querying the Hospital Database [12 Marks]

Write and execute the following SELECT queries. For each query, provide the **SQL statement** and a **screenshot of the result**:

- i. Retrieve all patients who live in **Lahore**.
- ii. List all doctors whose **experience is more than 5 years**, showing only their name, specialization, and fee.
- iii. Find all appointments that have status = '**Completed**'.
- iv. Show the **full name and age** of the **youngest patient** in the database (use MIN or ORDER BY LIMIT).
- v. List all appointments along with the **patient's full name and doctor's full name** (use JOIN).
- vi. Count the **total number of appointments per doctor** (GROUP BY with COUNT).

Note: You must use at least one JOIN query and one aggregate function (COUNT, MIN, MAX, AVG, SUM).

PART C: UPDATE & DELETE Operations [10 Marks]

Q4. Updating Records [5 Marks]

Perform the following UPDATE operations:

- i. A doctor has completed additional training. Update the **experience_years** of any one doctor by adding 2 years to their current value.
- ii. One patient's contact number has changed. Update the **contact_no** of any one patient.
- iii. Change the **status** of one 'Scheduled' appointment to 'Completed' and update the **diagnosis** field with a realistic diagnosis note (e.g., 'Hypertension – Stage 1').

Required Deliverables for each UPDATE:

- Write the UPDATE SQL statement.
- Screenshot BEFORE the update (SELECT showing old value).
- Screenshot AFTER the update (SELECT showing new value).

Q5. Deleting Records & Constraint Handling [5 Marks]

Perform the following DELETE operations:

- i. Delete an appointment that has status = '**Cancelled**'.
- ii. Try to delete a **doctor who has existing appointments**. Document what error you get and explain in 2–3 lines why this error occurs (relate to Foreign Key constraint).

Required Deliverables:

- SQL DELETE statement for each operation.
- Screenshot of the successful deletion result.
- Screenshot of the error (for part ii) and your written explanation.

PART D: Entity Relationship Diagram (ERD) [8 Marks]

Q6. ERD Design & Theory [8 Marks]

Based on the **City General Hospital** database you built above, draw a complete **Entity Relationship Diagram (ERD)**. Your ERD must be neatly drawn (on paper and scanned/photographed, or created digitally using draw.io / Lucidchart / dbdiagram.io) and must include:

- **All three entities:** Patient, Doctor, Appointment — each shown as a rectangle.
- **All attributes** of each entity — shown as ovals (or listed inside the rectangle for crow's foot notation).
- **Primary keys** — underlined in the ERD.
- **Foreign keys** — clearly indicated with arrows or relationship lines.
- **Relationships** — properly labeled (e.g., 'books', 'is treated by') with correct **cardinality notation** (One-to-Many, Many-to-One, etc.).

Theory Questions (answer in 3–5 lines each):

- What is the difference between a **Primary Key** and a **Foreign Key**? Give one real-world example of each from your hospital database.
- Explain what is meant by **Referential Integrity** and how it is enforced in your database.
- What is the difference between a **One-to-Many** and a **Many-to-Many** relationship? Can an Appointment table represent a Many-to-Many relationship between Patients and Doctors? Justify your answer.

Part	Topic	Marks
A	INSERT Operations (Table creation + data insertion)	10
B	SELECT Queries (Simple + JOIN + Aggregate)	12
C	UPDATE & DELETE + Constraint Handling	10
D	ERD Design + Theory Questions	8
	TOTAL	40

Good Luck! Work hard, be honest, and let your SQL do the talking. — Hafiz Muhammad Usman

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DATABASE SYSTEMS – QUIZ

Practical Problem Solving Quiz | Printed Submission

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Quiz Instructions:

1. This quiz is **practical and problem-based**. You must write and execute real SQL queries for each question.
2. For EVERY question: write the **SQL query** clearly and paste the **screenshot** of its execution and output.
3. Use any RDBMS (MySQL, MariaDB, SQLite). Use the database/scenario provided in each question.
4. **No MCQs**. All questions require written SQL and screenshots.
5. Plagiarism and AI-generated answers are strictly prohibited. Zero marks for copied work.
6. Write your **Name, Roll No., and Section** clearly at the top.

■ QUIZ SCENARIO – Online Bookstore

You are working as a junior Database Developer at **KitaabGhar – An Online Bookstore**. The company maintains a database with information about **Books, Authors, and Orders**. You are required to create this database, insert sample data, and perform SQL operations as instructed in each question below. All work must be supported by SQL queries and screenshots.

Table

Columns

books	book_id (PK), title, genre, price, stock_qty, author_id (FK), published_year
authors	author_id (PK), full_name, country, email
orders	order_id (PK), book_id (FK), customer_name, quantity, order_date, total_amount

Q1. Database Setup & Data Insertion [10 Marks]

Set up the KitaabGhar database as follows:

- Create the database **KitaabGhar_DB** and all three tables with correct data types, PRIMARY KEY on each table, and FOREIGN KEY constraints (book references author; order references book).
- Insert at least **5 authors** (from at least 3 different countries).
- Insert at least **8 books** (mix of genres: Fiction, Non-Fiction, Science, History). Make sure stock_qty varies — some books should have low stock (e.g., 2–5 copies).
- Insert at least **6 orders** with different customers, books, and dates.

Deliverables:

- All CREATE TABLE and INSERT INTO statements.
- Screenshot: SHOW TABLES; result.
- Screenshot: SELECT * FROM each table showing all data.

Q2. SELECT Queries – Real World Retrieval [14 Marks]

Write and execute the following SELECT queries (SQL + screenshot for each):

- Retrieve all books where **genre = 'Fiction'** and **price < 800**.
- List all books along with their **author's full name and country** (use JOIN).
- Find the **most expensive book** in the store (use MAX or ORDER BY LIMIT 1).
- Show the **total revenue generated per book** (use GROUP BY on orders, SUM of total_amount).
- List all orders placed in the **month of April 2025** (use MONTH() or BETWEEN on order_date).

Hint: You may need to insert order_date values in '2025-04-xx' format in your test data.

Q3. UPDATE – Modifying Real Records [10 Marks]

Perform the following updates in your database (SQL + before/after screenshots):

- The price of a book titled **'The Alchemist'** (or any book you inserted) has been revised. **Increase its price by 15%** using a single UPDATE statement.
- A customer complained. **Reduce the stock_qty** of the book with the highest number of orders by 3.
- One author changed their **email address**. Update it.

For each UPDATE: Show the SELECT result before and after the change.

Q4. DELETE & Constraint Analysis [6 Marks]

Perform the following operations and document the results:

- i. Delete all orders where **quantity = 1** (low-volume orders).
- ii. Now try to **delete an author** who still has books in the books table. Take a screenshot of the error. Then explain in 3–4 lines: *What is a Foreign Key Constraint Violation and why does the database prevent this deletion?*

Deliverables:

- DELETE SQL statements.
- Screenshot of successful deletion result.
- Screenshot of the error (for part ii).
- Written explanation (3–4 lines) of the FK constraint violation.

Q#	Topic	Marks
Q1	Database Setup + INSERT (Tables + Data)	10
Q2	SELECT Queries (JOIN, Aggregate, Filters)	14
Q3	UPDATE Operations (3 scenarios)	10
Q4	DELETE + FK Constraint Analysis	6
	TOTAL	40

Be honest. Be practical. Be a real database developer. — Hafiz Muhammad Usman